

Text mining by using Python: Application to patent documents

(2025 Project description)

Instructor: Kazuyuki Motohashi (motohashi@tmi.t.u-tokyo.ac.jp) and his lab members

1. Introduction

This project is for understanding practices of applying NLP techniques for technology management analysis using patent text information. The project starts with a course work for one week with some preparatory activities, followed by individual project works (mainly topic modeling of some particular type of technology) for some weeks.

2. Setting

- PC with Python: You need to install Python 3.X “anaconda” environment. We use “Jupyter notebook” inside “anaconda” for this project. It can be downloaded freely.
- I will let you access to the shared drive of Google Suite of my laboratory (motolabo.net). Python_Text_Mining holder contains
 - Some files used in this project (including this file)
 - Coursera_video directory: Michigan U, “Applied Text Mining by Python”, referred as “VIDEO” here after
 - Data for topic modeling project
- Please check whether you could access to the following holder. If it is not the case, report to prof. Motohashi with your name and e-mail address.
https://drive.google.com/drive/folders/1eIbsqmaCnY8tjryYRnriZ2Wu_stQl7g9?usp=sharing

The folder of “Internahip_files” is only for those who did the phase 2 of this project, “innovation discovery from patent and web contents information”

3. Preparatory activities (Will be introduced at the orientation day)

- Preparatory activities to start course work is provided in the appendix 1.

4. Whole structure

- Part 1 : June 9- June 13 : course works with practical exercise
- Part 2 : June 16 – August 1 : internship project with real data

[Phase 1]: Learning text mining technique by small sample data : June 9 – June 13
(Orientation will take place on June 2, as follows)

Venue : Zoom Conference Room : <https://u-tokyo-ac-jp.zoom.us/j/99083411670?pwd=U0dicHlONXNiUjBmUFU3Y1h6NWFZdz09>

Orientation Day: June 2, 10:00-12:00 Thai& Vietnam Time (12:00-14:00, JST)

- Explanation of home works etc (Appendix 1)
- Web crawling introduction by Suchit

Day1: June 9, 10:00-12:00 Thai& Vietnam Time (12:00-14:00, JST)

- Review of preparatory activities
- How to convert technology contents info (in patent document) to numeric (vector) information? : Keyword extraction, TF-IDF, Word/Document embedding (such as Word2Vec)
- HOMEWORK1: Extract three keywords by using TF-IDF score from 25 AI patents abstracts, supplied as a home work file. TF-IDF score can be obtained easily by Python module like “scikit-learn” and “genism”. But I would advice you to do it without such modules for your understanding of the concept. There are many hints over internet such as <https://towardsdatascience.com/text-summarization-using-tf-idf-e64a0644ace3>

Day 2: June 11, 10:00-12:00 Thai& Vietnam Time (12:00-14:00, JST)

- Review of homework of Day 1
- What are ML applications of patent text information?
- Preprocessing of text: tokenization regularization (lowering character), stemming/lemmatization and stop words exclusion
- Text processing of tf-idf vectors:
 - ① Create dictionary: mapping every word to a number
 - ② Corpus (list of bags of words) : a list of number of words occurring in each documents
- HOMEWORK2: Calculation of similarity measures across each documents, and find out the closed documents of each of 25 AI patents : Now please use genism module, again there are many hints over internet such as “How do I compare document similarity using Python?” by Jonathan Mugan

<https://www.oreilly.com/content/how-do-i-compare-document-similarity-using-python/>

Day 3: June 13 10:00-12:00 Thai& Vietnam Time (12:00-14:00, JST)

- Review of homework of Day2
- Understanding **technology trend by patent information (clustering vs topic modeling)**
- Watch VIDEO 16&17
 - ① Understanding the concept of topic modeling
 - ② LDA model by using genism module
- ~~Introduction for your **project work** : Pick up one of subject areas shown in the appendix 2 to construct LDA topic model, then understand the contents of technological development of that field. More instructions will be provided later with individual consultation of works to be done.~~
- Some references
 - ① There is a very easy to read document of LDA model available in the shared drive. (Beginners_Guide_Topic_Modeling.doc)
 - ② Good reference about topic modeling by genism
<https://www.machinelearningplus.com/nlp/topic-modeling-gensim-python/>

[Phase 2]: Project work with **final presentation : June 16 – August 1**

Orientation Day: June 16, 10:00-12:00 Thai& Vietnam Time (12:00-14:00, JST)

- Orientation of phase 2
- **Dual Attention Model**

(Outline of works, details to be announced)

- This process is using NLP techniques above to real world exercises, such as **technology opportunity discovery tasks**. Please refer to the following Youtube video for this research project.
<https://www.youtube.com/watch?v=Y6TIIHxfKsmM&t=581s>
- Major tasks are scraping company's web pages and using such text information together with patent information to predict market opportunities based on particular type of technology.
- Web scraping task will **start after June 2 orientation** (with a short course of web scraping)
- Some research assistant honorarium will be offered to this works.

Appendix1 Preparatory Homework

1. Python environment and understanding on basic command

- You need to bring your own PC with Python 3.X anaconda environment, particularly Jupyter notebook (download for free)
- Familiarize yourself with basic Python function dealing with text, such as making word list from sentence, word count and distribution
- Playing with NLTK library of Python

2. Tasks

- Watch **Video 01,05 and 06** in “Coursera_Video” directory) for the following contents
 - What is National Language Processing?
 - Playing with NLTK (Natural Language Processing Tool Kit)
- Use the **patent abstract data** (25 AI related patent abstracts, provided as “homework.csv”), to calculate
 - How many sentences, words, unique words
 - List of **words occurring no less than 10 times and no less than 5 character lengths**
 - After regularization (lowering characters), then **comparing the case of Porter Stemmer and Lenmatizer**. Compare the results and which one would you think it is better and why?
 - Any other observations?

3. Notes

- You may go ahead to **watching other videos**, but I will give you some general directions.
 - **VIDEO 02 and 03** are for regular expression, working with Pandas module of python (concept of “data framework”) : No need to worry about them for now, but it will be useful for **rule based text manipulations**.
 - **VIDEO 04** is for **character code information for various languages** (Python’s default character set is UTF-8, no need to worry about this for now either.
 - Watching **VIDEO 07 is optional** (could be helpful for your subsequent tasks)

